UNIVERSITY AT ALBANY
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College of Arts and Sciences
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Colloquium

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Applications of Spectral Graph Theory and Gaussian Process Regression to Problems in Genetics

Friday, October 23, 2026
3:00 p.m. Massry B010
(tea & coffee at 2:40 p.m.)

Abstract. Many important biological molecules, including DNA, RNA, and proteins, can be represented as strings of fixed length over a finite alphabet. Because mutations often change just one letter at a time, it is natural to represent the possible sequences as the vertices of a Hamming graph, with edges connecting sequences that differ at a single position and measurable properties of these molecules represented as functions on the vertices. Recent advances in high-throughput molecular assays now make it possible to measure such properties for tens of thousands to millions of sequences in a single experiment. However, because the number of vertices in a Hamming graph grows exponentially with sequence length, these measurements often cover only a small fraction of its vertices. In this talk, I will discuss spectral approaches to analyzing these data and reconstructing functions on Hamming graphs. I will also describe methods for Gaussian process regression on these graphs inspired by this spectral perspective, including GPU-accelerated methods for working with longer sequences. Finally, I will consider how to understand and visualize the qualitative structure of the inferred functions

on graphs containing up to millions of vertices. I will describe an approach to nonlinear dimensionality reduction that uses models from population genetics to define evolutionary random walks on the Hamming graph, and then derives embedding coordinates from the subdominant eigenvectors of the resulting rate matrix. Together, these approaches illustrate how the structure of Hamming graphs can guide the development of computational and statistical tools for understanding genetics and evolution.